EDUSPARK ANALYTICS · INTELLIGENCE PLATFORM

Customer Segmentation & *A/B Testing* Suite

50

YEARS OF DATA HERITAGE

Personalized learning interventions driven by ML clustering, behavioral scoring, and controlled experimentation across 12,400+ schools in 47 districts. Grades 8–12.

ENTITY RELATIONSHIP MODEL — LINKED DATA ARCHITECTURE

students
Core entity · fact table

student_id	uuid PK
school_id	uuid FK
cohort_id	uuid FK
grade	int(8..12)
age	int
gender_code	char(1)
first_enrolled	date
ell_status	bool
iep_flag	bool
free_lunch	bool

assessment_scores
Measurement · time-series

score_id	uuid PK
student_id	uuid FK
test_date	date
subject	enum
raw_score	float
percentile	float
grade_equiv	float
geo_score	float
stanine	int(1..9)
is_makeup	bool

schools
Dimension · geo context

school_id	uuid PK
district_id	uuid FK
school_name	text
school_type	enum
latitude	float
longitude	float
locale_code	enum
enrollment	int
title1_status	bool
avg_class_size	float

districts
Dimension · policy level

district_id	uuid PK
district_name	text
state_code	char(2)
region	enum
funding_tier	int(1..5)
urbanicity	enum
tech_adoption	float
contract_start	date
arn	text
seats_licensed	int

ab_assignments
Experiment · derived table

assignment_id	uuid PK
student_id	uuid FK
experiment_id	uuid FK
variant	enum
assigned_at	timestamp
segment_label	char(1)
pre_score	float
post_score	float
delta_score	float
converted	bool

Links via student_id → students and experiment_id → experiments

students ↔ assessment_scores (1:N)

students → schools → districts (N:1:1)

students ↔ ab_assignments (1:N)

assessment_scores ⊕ districts = ml_feature_matrix

ML PIPELINE — FROM RAW SCORES TO SEGMENTS

01
Feature Engineering
Rolling 3-year score deltas, subject strength ratios, geo-index normalization, socioeconomic proxy scores
`Pandas + dbt`

→

02
Dimensionality Reduction
PCA (12 → 4 components, 89% variance retained) + UMAP for cluster visualization embedding
`scikit-learn`

→

03
Clustering
K-Means (k=5 by elbow + silhouette), DBSCAN for outlier cohorts, hierarchical for district roll-up
`k=5 · sil=0.71`

→

04
Segment Scoring
Random Forest classifier trained on labeled cohorts (F1=0.88), LTV regression per segment, churn propensity
`RF · XGBoost`

→

05
Experiment Design
Power analysis per segment (α=0.05, β=0.20), stratified random assignment by school & grade
`Bayesian + freq.`

5 IDENTIFIED CUSTOMER SEGMENTS — K-MEANS · N = 2.1M STUDENTS

SEG A
🌟 Rising Stars
High-trajectory students in underfunded districts. Strong math, developing English. Grade 9–11.

Students	683K
Avg score Δ/yr	+8.4%
Geo index	0.38 (Rural)
LTV score	87/100

SEG B
🏆 Suburban Achievers
Well-resourced, consistent performers. High science scores. Grades 10–12. Parent engagement high.

Students	502K
Avg score Δ/yr	+4.1%
Geo index	0.78 (Suburban)
LTV score	94/100

SEG C
🔧 At-Risk Rebuilders
Declining trajectories, high absenteeism proxy. Need scaffolded support. Grades 8–10 concentrated.

Students	355K
Avg score Δ/yr	-2.1%
Geo index	0.22 (Urban)
LTV score	52/100

SEG D
🔬 STEM Specialists
Extreme math/science advantage, below-avg English. Gifted programs. Grade 11–12, urban elite schools.

Students	218K
Avg score Δ/yr	+11.2%
Geo index	0.91 (Metro)
LTV score	91/100

SEG E
🌐 ELL Emerging
English Language Learners. High geo-variability. Low English, moderate math. Fastest-growing cohort.

Students	142K
Avg score Δ/yr	+6.8%
Geo index	0.44 (Mixed)
LTV score	73/100

ACTIVE & COMPLETED A/B EXPERIMENTS

Adaptive Hint Frequency
• Live

"More frequent micro-hints increase math score velocity in Seg A"

CONTROL · A	TREATMENT · B
Standard hints on error	Proactive micro-hints

Geo-Contextualized Problems
✓ Completed

"Locally-anchored science scenarios improve engagement in Seg E"

CONTROL · A	TREATMENT · B

Score Δ	+3.2%	Score Δ	+7.8%
Completion	71%	Completion	83%
n	12,440	n	12,381

+143% relative lift · p < 0.001 · **95% CI [3.8, 6.4]**

Generic problems		Geo-contextualized	
Engagement	58%	Engagement	74%
Retention D30	44%	Retention D30	61%
n	8,220	n	8,195

+38% D30 retention lift · p = 0.004 · **Rolled out to Seg E**

Difficulty Auto-Scaling (DAS)

• Live

"IRT-based adaptive difficulty reduces frustration in Seg C"

CONTROL · A		TREATMENT · B	
Fixed curriculum path		IRT adaptive path	
Dropout rate	31%	Dropout rate	18%
Score Δ	-1.8%	Score Δ	+2.4%
n	9,110	n	9,084

-42% dropout reduction · p = 0.018 · **Bayesian: 97.2% prob. superior**

Teacher Dashboard Nudges

⏳ Planned Q3

"Weekly AI-generated insight nudges lift district renewal rates"

CONTROL · A		TREATMENT · B	
Standard reports		AI insight nudges	
Renewal est.	78%	Renewal est.	85%+
Login freq.	1.2×/wk	Login freq.	2.8×/wk
n (schools)	600	n (schools)	600

+7pp renewal target · Power: 85% · **Launching Sep 2025**

Avg subject scores by segment

	A	B	C	D	E
English	72	88	58	66	49
Math	85	84	61	96	74
Science	78	92	55	98	68
Geo	64	81	48	73	71

Score trajectory vs geo-index (UMAP projection)

Score trajectory

← Rural · Geo Index · Metro →

• Seg A • Seg B • Seg C • Seg D • Seg E

A	B	C	D	E

DISTRICT × SUBJECT PERFORMANCE HEATMAP — TOP 10 DISTRICTS

	Eng	Math	Sci	Geo	8th	9th	10th	11th	12th	Avg
Riverside USD	87	65	87	58	67	60	61	60	72	**69**
Oakland Unified	58	98	90	60	92	72	64	75	77	**76**
Mesa ISD	79	61	78	62	72	87	71	58	93	**73**
Fulton County	96	92	76	57	61	78	68	88	91	**79**
Denver PS	75	92	96	70	63	98	77	83	72	**81**
Atlanta City	44	52	40	44	53	81	74	84	80	**61**
Portland SD	59	49	77	61	67	43	73	57	45	**59**
Phoenix UHD	54	71	66	60	67	51	63	80	77	**65**
Boston PS	82	42	69	69	78	84	79	43	52	**66**
Clark County	48	75	67	50	43	49	56	55	81	**58**

RESULTING OUTCOMES — POST-SEGMENTATION DEPLOYMENT

📈

+23%

Mean score improvement (Seg A+C combined, 18-month cohort)

↑ vs +9% baseline pre-segmentation

🔄

91%

District contract renewal rate after DAS rollout

↑ from 78% the prior year

⚡

3.4×

Faster intervention targeting via ML segment classifier

↑ 12 hrs → 3.5 hrs per counselor cycle

💰

$4.2M

Incremental ARR from upsell to Seg B & D schools

↑ ROI on A/B program: 11.4× in Y1